Introduction to Web Programming

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Introduction

Today's Topics

- Internet
- World Wide Web
- Web Architecture
- Common Gateway Interface
- CGI Programming

Internet

- A world-wide network of **computer networks**.
- **All connections use TCP/IP** (Transport Control Protocol/Internet Protocol)
- TCP/IP hides the differences among devices connected to the Internet
- Internet provides several protocols: **Telnet(DOS Prompt), FTP, mailto, HTTP, etc.** All with different interfaces. A solution is the World Wide Web (WWW)

World Wide Web

- Allows users from anywhere on the internet to **retrieve documents** from any other computers
- The main protocol is the **HTTP** - HyperText Transfer Protocol
- HTTP provides a standard form of **communication between** web browsers and web servers.
- Web or Internet?
  - *Internet* is a collection of computers.
  - *Web* is a collection of software and protocols
    - a set of HTML pages accessible using the HTTP protocol

The World Wide Web (History)

- 1989 – WWW concept

  by Tim Berners-Lee

  - Scientist at CERN (European centre for High Energy Physics – Geneva)
  - Hypertext Markup Language (HTML)



The World Wide Web (History)

- 1989 - Birth of www
  - Tim Berners-Lee & his associates at CERN
- 1990 - First Web Browser/Editor Program
  - Used within CERN
- 1991 - Public offering of www
- 1993 - Birth of Mosaic
  - Graphical, multimedia browser from NCSA
- 1994 - First commercial browser
  - By Netscape communications founded by Jim Clark and Marc Andreessen

The World Wide Web (History)

- Some facts
  - 1994 – Hotmail starts web based email
  - 1994 – World Wide Web Consortium (W3C) was founded
  - 1995 – JAVA source code was released
  - 1996 – Mirabilis (Israel) starts ICQ
  - 1998 – Google is founded

The Simplified WWW

Web browser wants a page from a **web server**

Pages structured into **HTML** and referenced via **URL**s

HTTP used to structure communication between client and server

This is the World Wide Web

get bytes (in this case HTTP packets) from users program from one machine to another

route groups of bytes (IP PACKET) between networks

(independent of hardware)

Protocol Use behind

The internet TCP, UDP and IP

get groups of bytes (FRAME) from A to B on a single Network

(hardware dependant)

get bits from A to B on a single network

(hardware dependant)

Existing network technologies

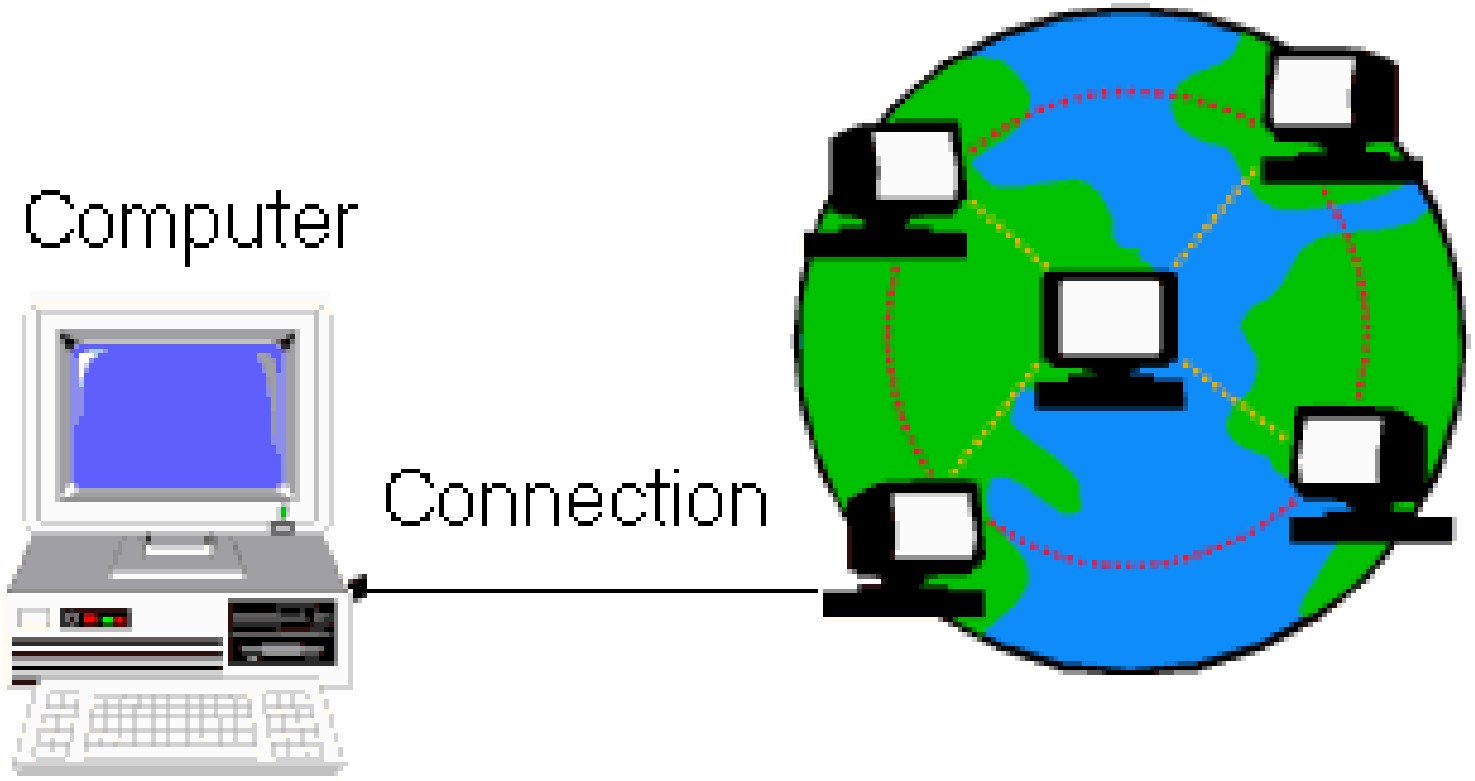
World Wide Web (cont.)

- Web browsers
  - Programs running **on client computers**
  - **Request** for documents using **URL**
  - *eg. MS Internet Explorer, Mozilla Firefox, Google Chrome etc.*

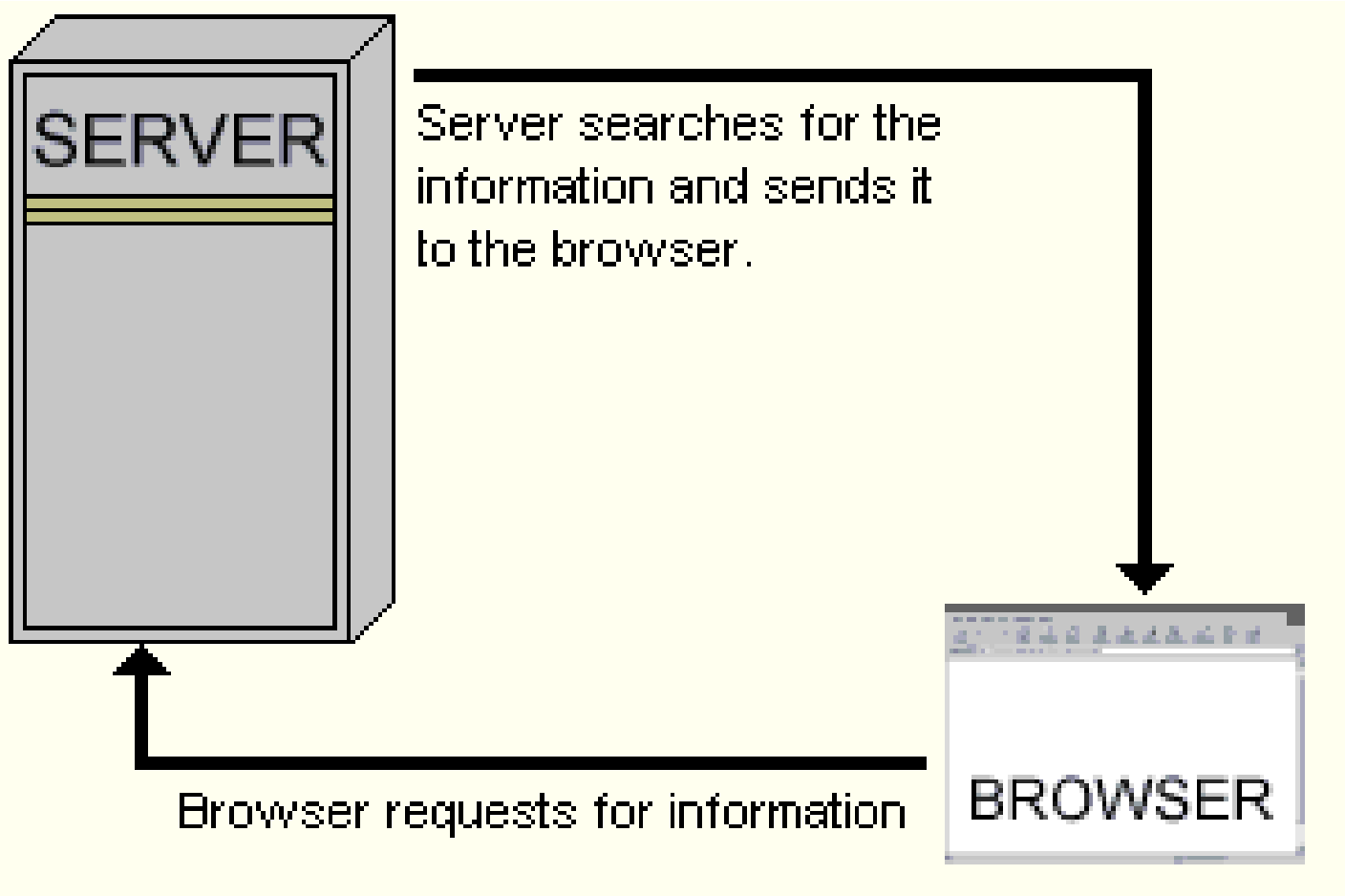
- Web servers
  - Programs that **provide documents to browsers**
  - *eg. Apache,*
    IIS(Internet Information Services) by Microsoft

- URL – Uniform Resource Locator
  - A **reference** of a document on the web.
  - *eg. http://comp.fsksm.utm.my/index.html*
  - *http://comp.fsksm.utm.my/images/logo.jpg*

Web Client (browser)



Web Server


SERVER
Server searches for the information and sends it to the browser.
BROWSER
Browser requests for information

URL

- Abbreviation of ***U**niform **R**esource **L**ocator,* the global address of documents and other resources on the World Wide Web.
- The first part of the address indicates what protocol to use, and the second part specifies the IP address or the domain name where the resource is located.
- For example, the two URLs below point to two different files at the domain *pcwebopedia.com*. The first specifies an executable file that should be fetched using the FTP protocol; the second specifies a Web page that should be fetched using the HTTP protocol:
  - **ftp://www.pcwebopedia.com/stuff.exe**
  - **ftp://www.pcwebopedia.com/stuff.zip**
  - **http://www.pcwebopedia.com/index.html**

URL

<protocol> <host name> <path>

- **HTTP**--Hypertext Transfer Protocol
  - Protocol for accessing World Wide Web documents
- **FTP**--File Transfer Protocol
  - Protocol for transferring files from one computer to another
- **Gopher**
  - **Protocol for accessing documents via Gopher menus (archaic; no longer widely used)**
- **Telnet**
  - **Protocol that allows users to logon to a remote computer**

URL Host Name

<protocol> <host name> <path>



<host>.<organization>.<org type>

www.utm.my

www.google.com

www.acm.org

www.jpa.gov.my

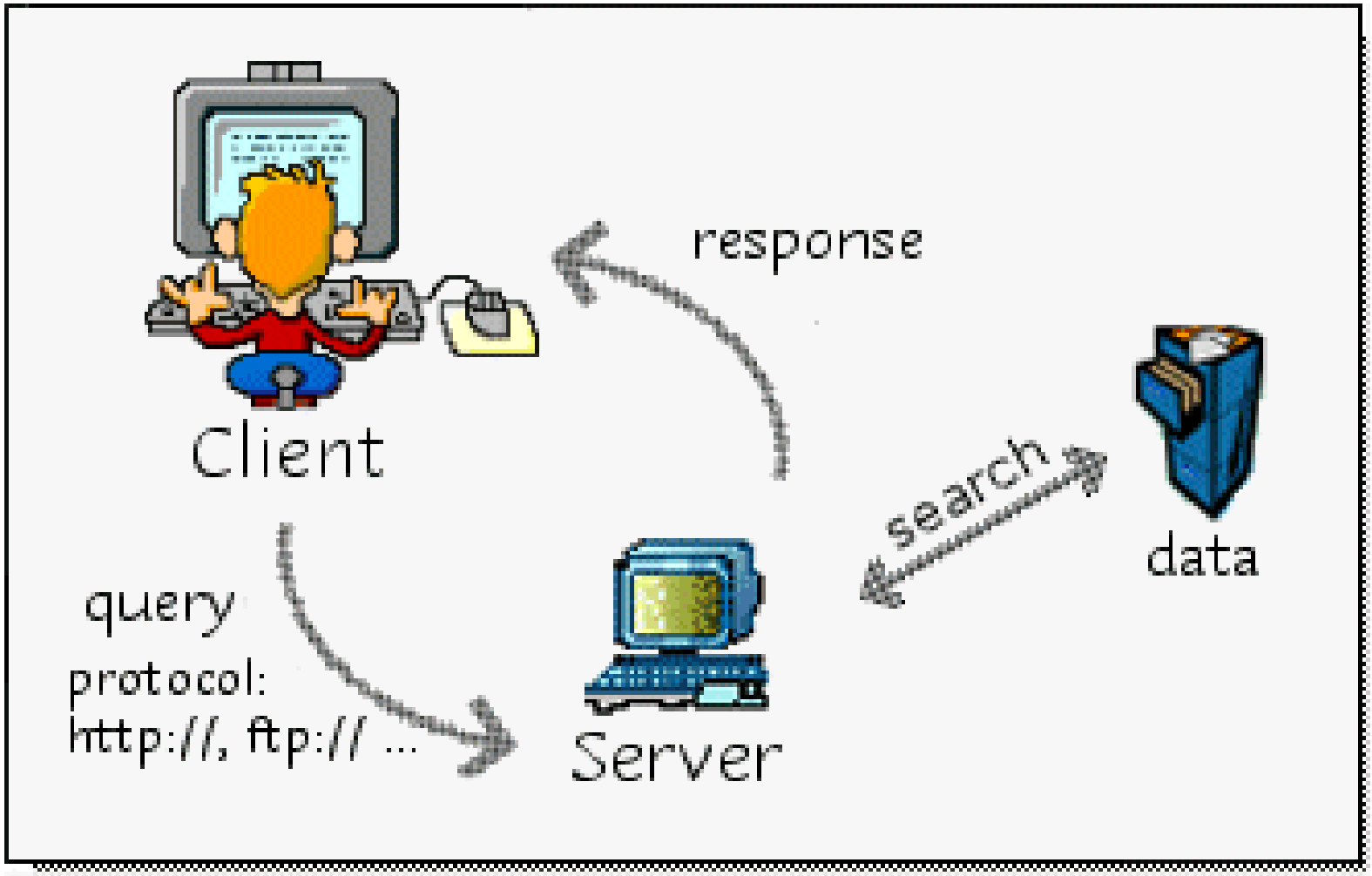
Organization Types

- **.org :** non-profit organizations
- **.com:** commercial hosts
- **.net :** network hosts
- **.edu :** educational institutions
- **.int :** international treaty organizations
- **.gov :** government agencies and organizations
- **.mil :** U. S. military
- **.my :** Malaysia
- **.jp :** Japan

URL Examples

- **Full URL**
  - **http://www.utm.my/**
  - **http://elearning.utm.my/**
  - **http://www.maxis.net.my/**
  - **http://www.lhdn.gov.my/**

Web Surfing (Logical View)



Web Architecture

- How web pages are **delivered** to us?
- 2 approaches:
  - **Static** content
  - **Dynamic** content

Static VS Dynamic

- HTML documents are usually **static**
- The contents can only be **changed manually**
- There are needs for **dynamic** documents to:
  - **Search results**
  - **Database access**
  - **Context sensitive reply**

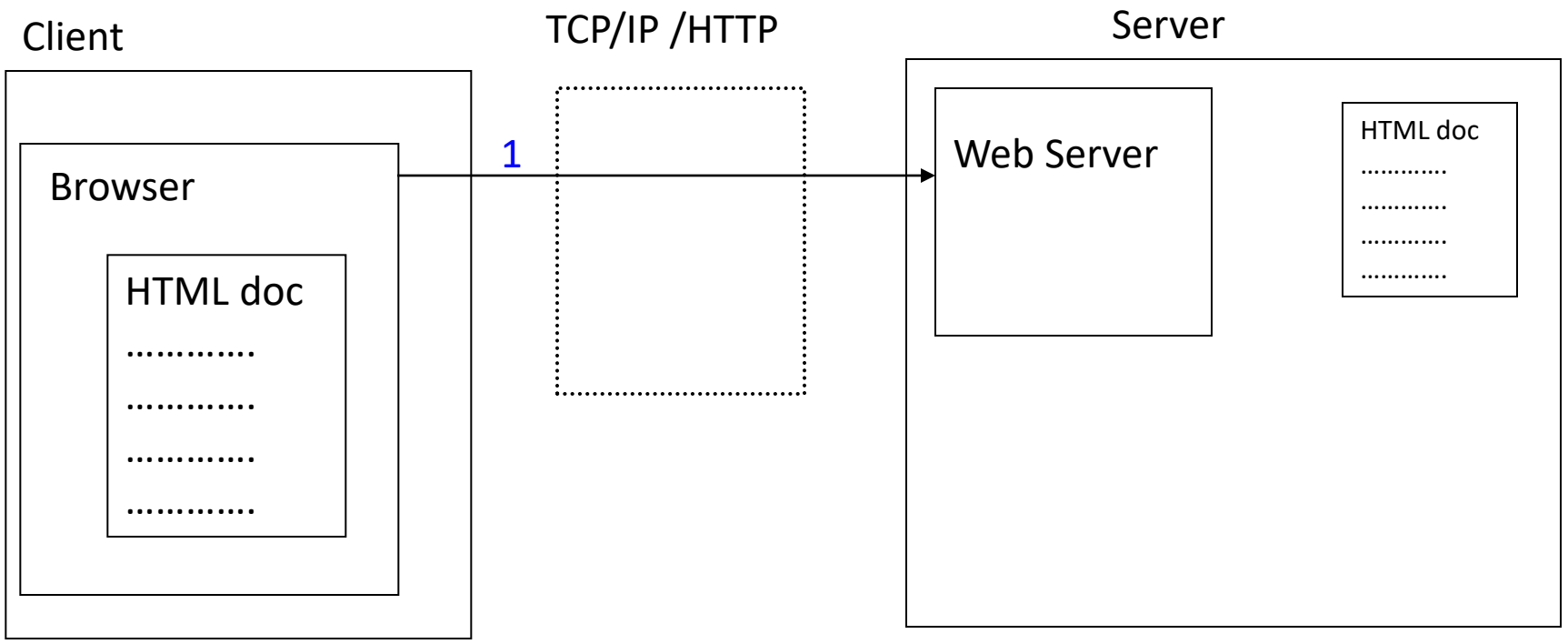
- Static
  - page appears **exactly as it was encoded**, nothing changes

- Dynamic
  - page is compiled, **or able to be changed**

Dynamic Web pages

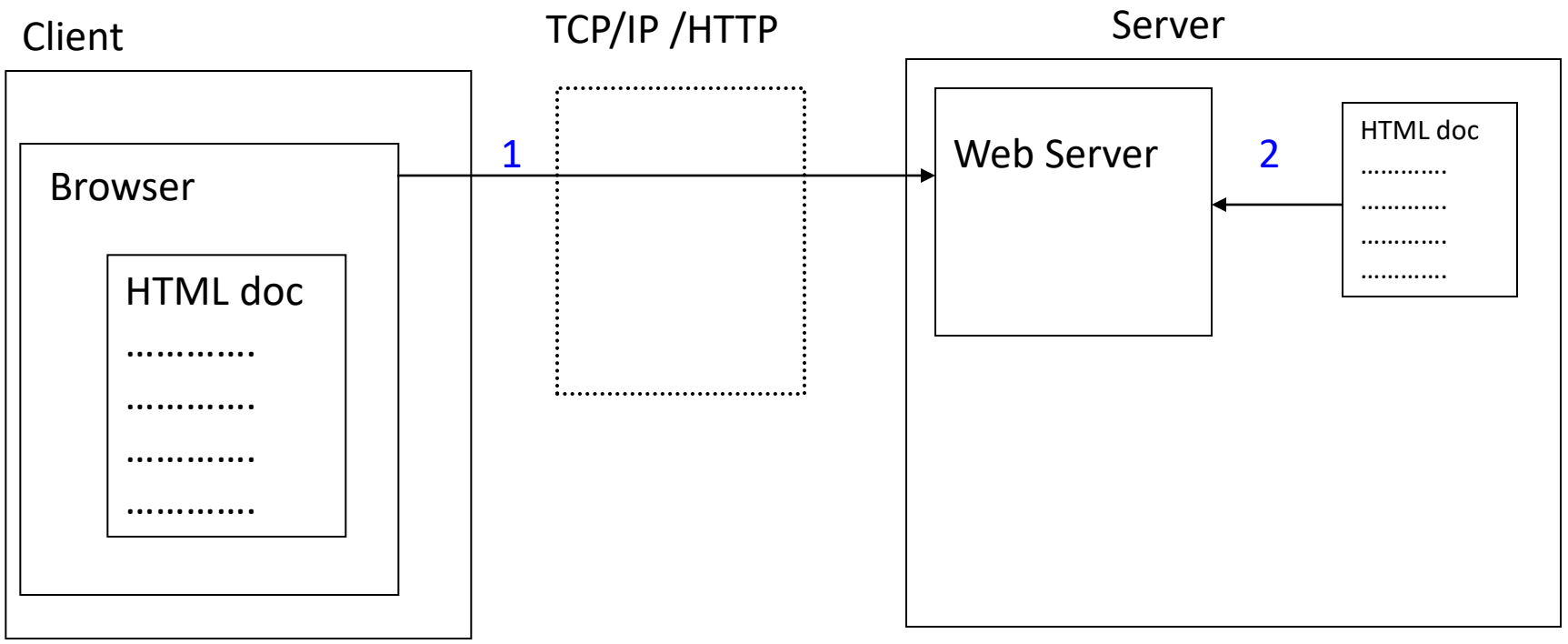
- Applications executed by the server at run-time **to process client input** or **generate document in response to client request**
- Generating dynamic Web pages requires **programming**

Web Architecture: Static content



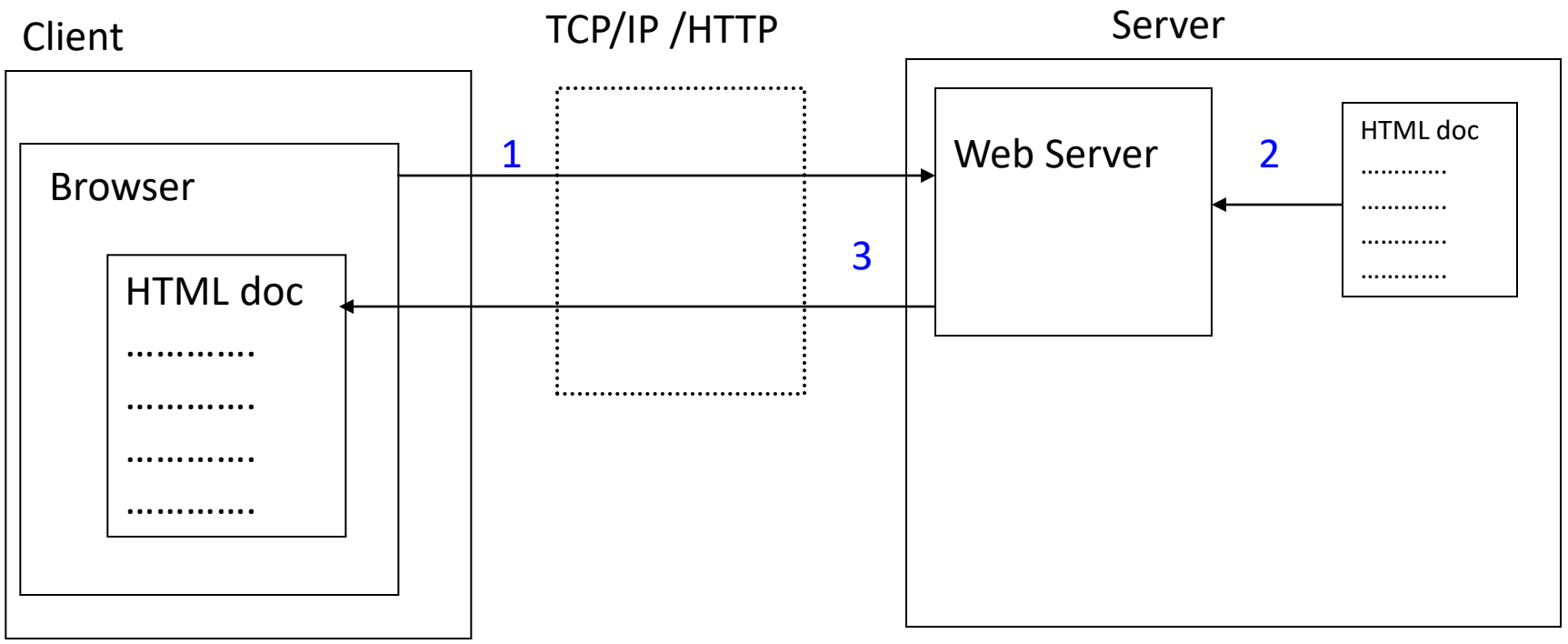
1. Client browser send a request for HTML document to web server

Web Architecture: Static content



2. Web server try to find and read the document at server directory

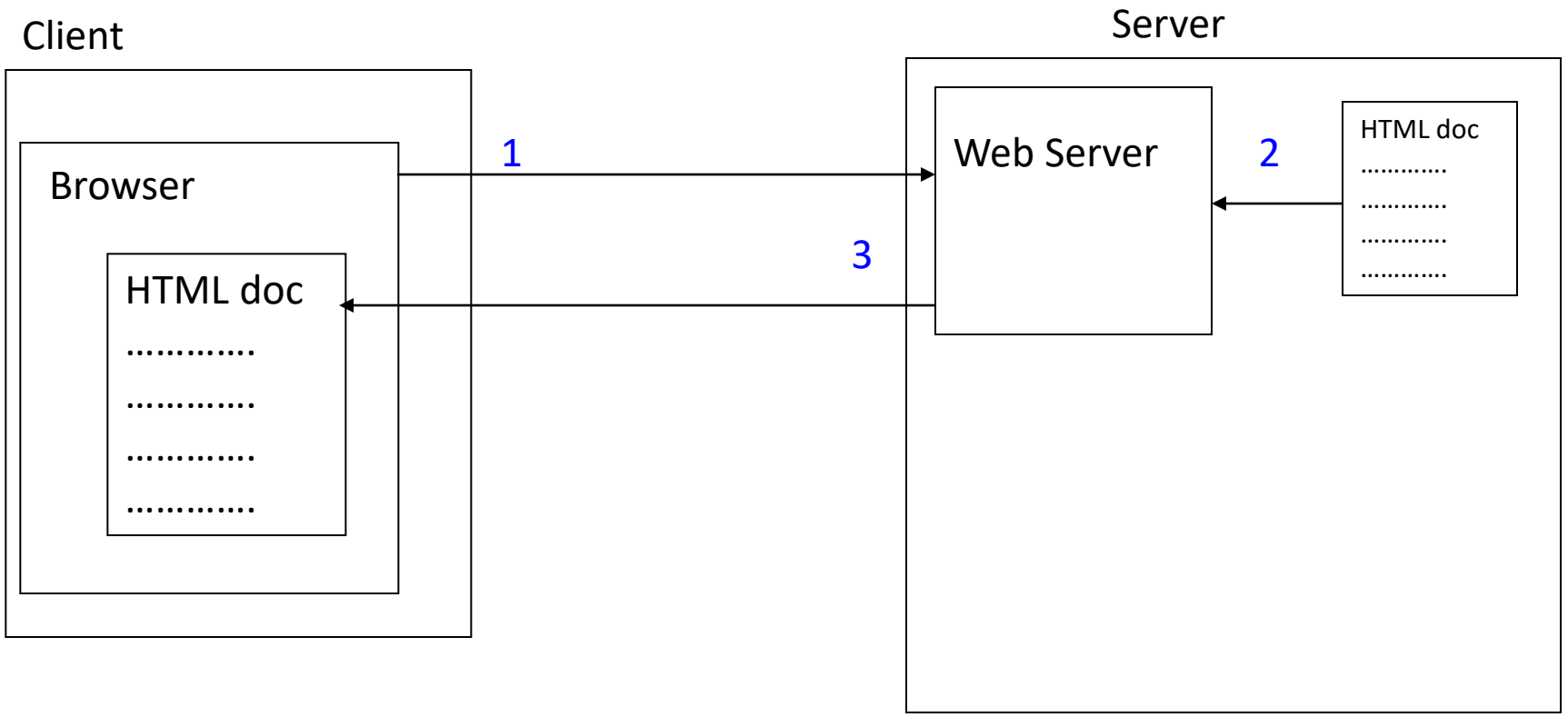
Web Architecture: Static content



3. Web server send the HTML document to the client browser

Web Architecture: Static content

How the Web works for static contents (basic architecture)



Web Architecture

- Dynamic contents :
  - CGI (Common Gateway Interface)
  - Server pages

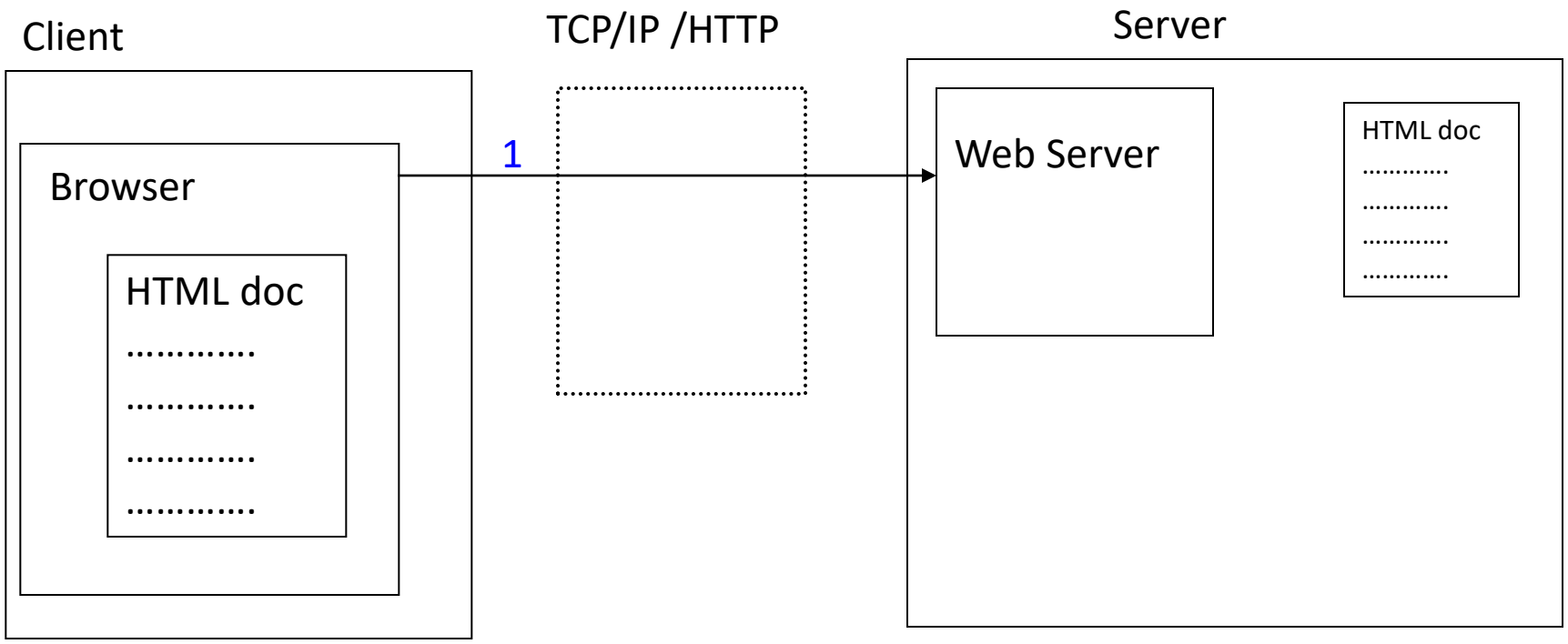
CGI (Common Gateway Interface)

- A CGI program is a program that follows the CGI specifications.
- The main role of **CGI programs is producing documents in HTML format** – ie. the only language that the web browsers can understand.
- Eg. getting data from a database,
  - A CGI program **submits SQL queries to the database server.**
  - After getting the result, the program formats it into HTML, then **give it to the web server**.
  - The **web servers then passes the HTML** document to **web browsers.**

Common Gateway Interface

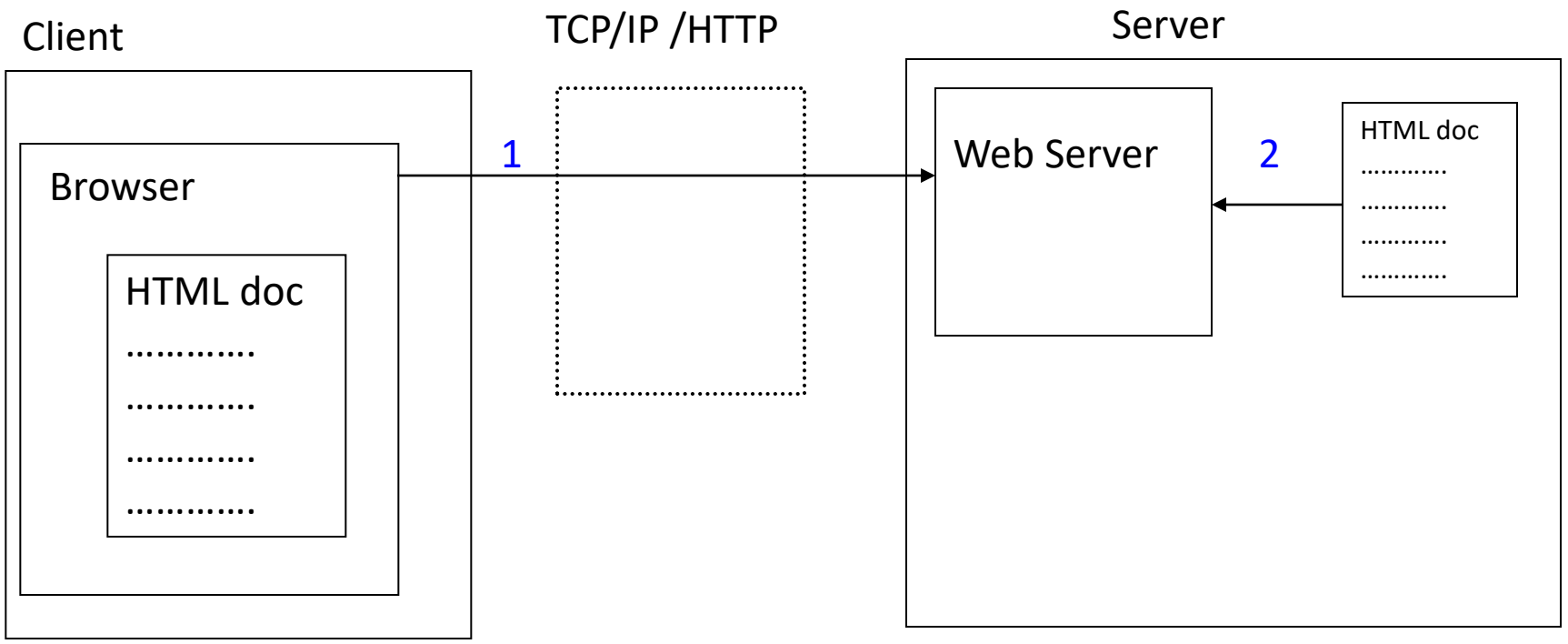
How it works ???

Common Gateway Interface



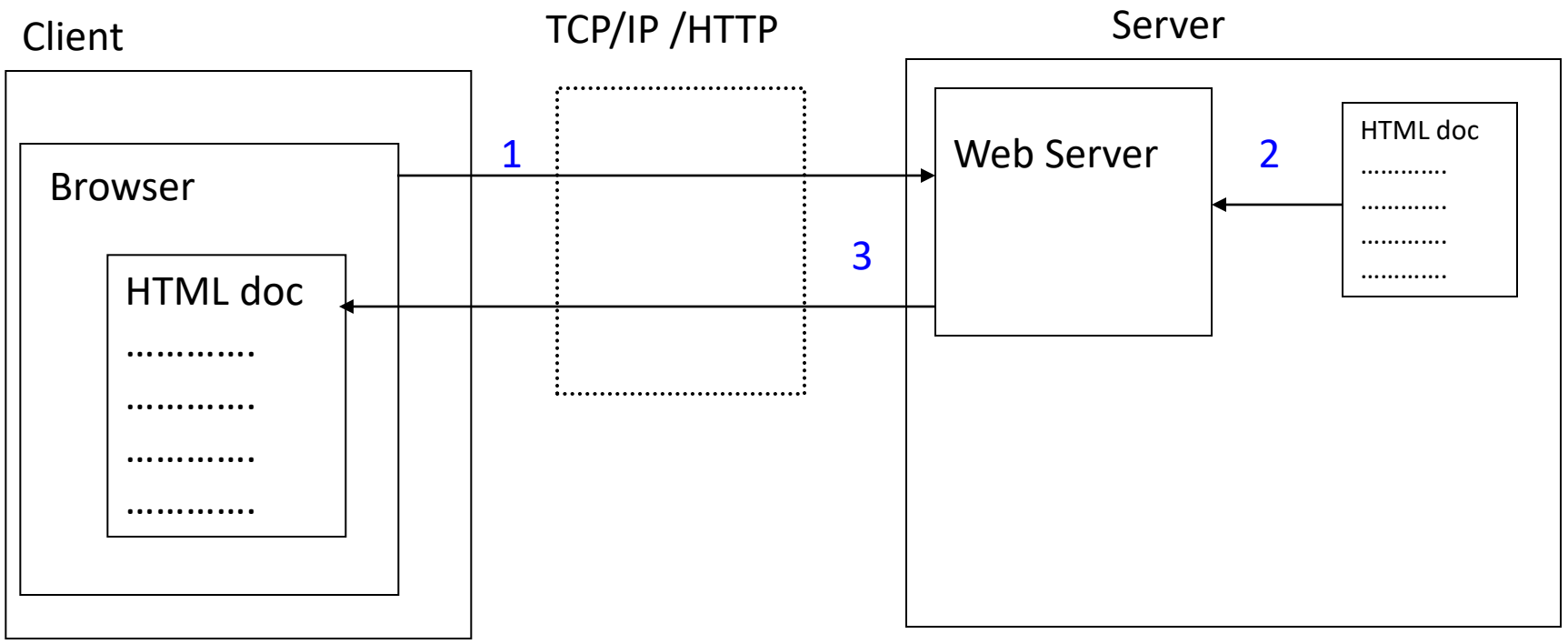
1. Client browser send a request for HTML document to web server

Common Gateway Interface



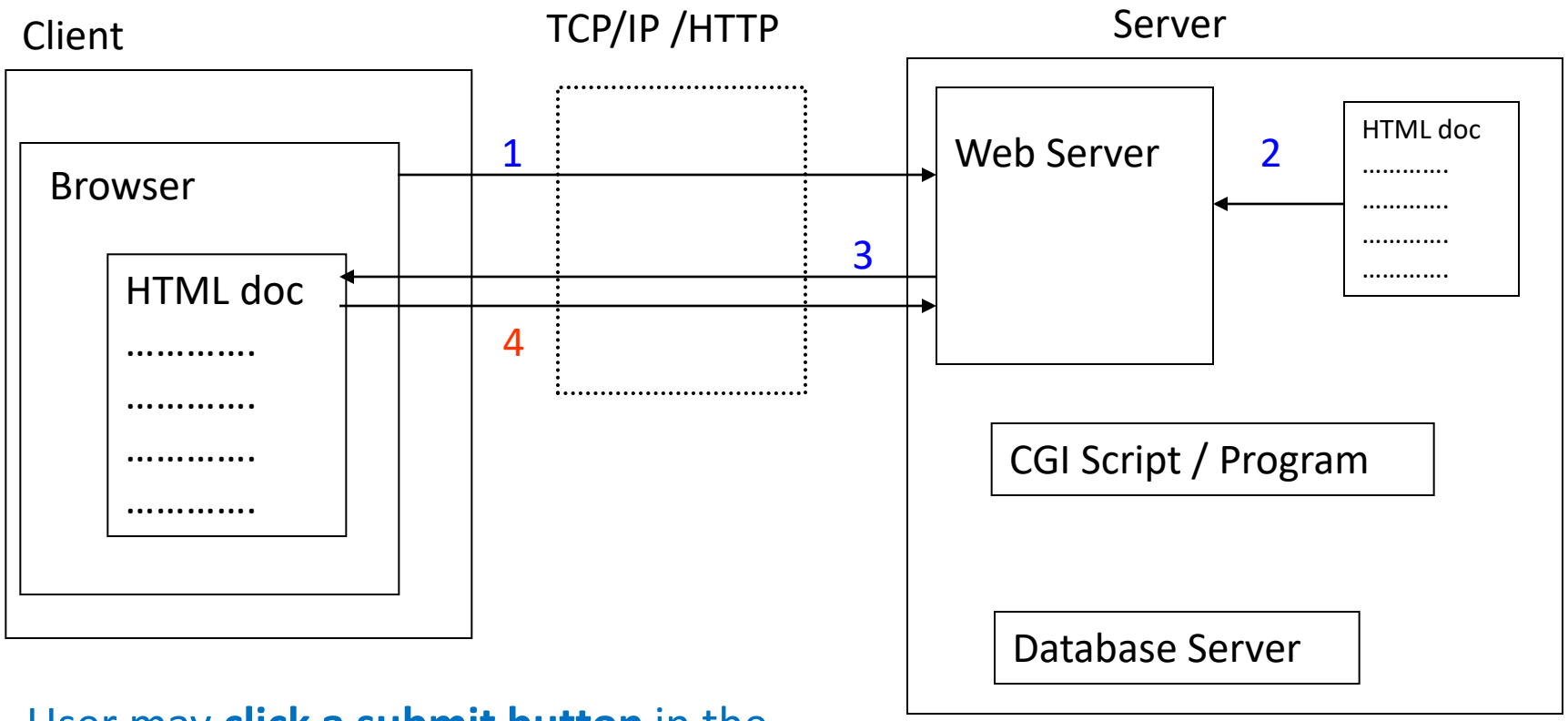
2. Web server try to find and read the document at server directory

Common Gateway Interface



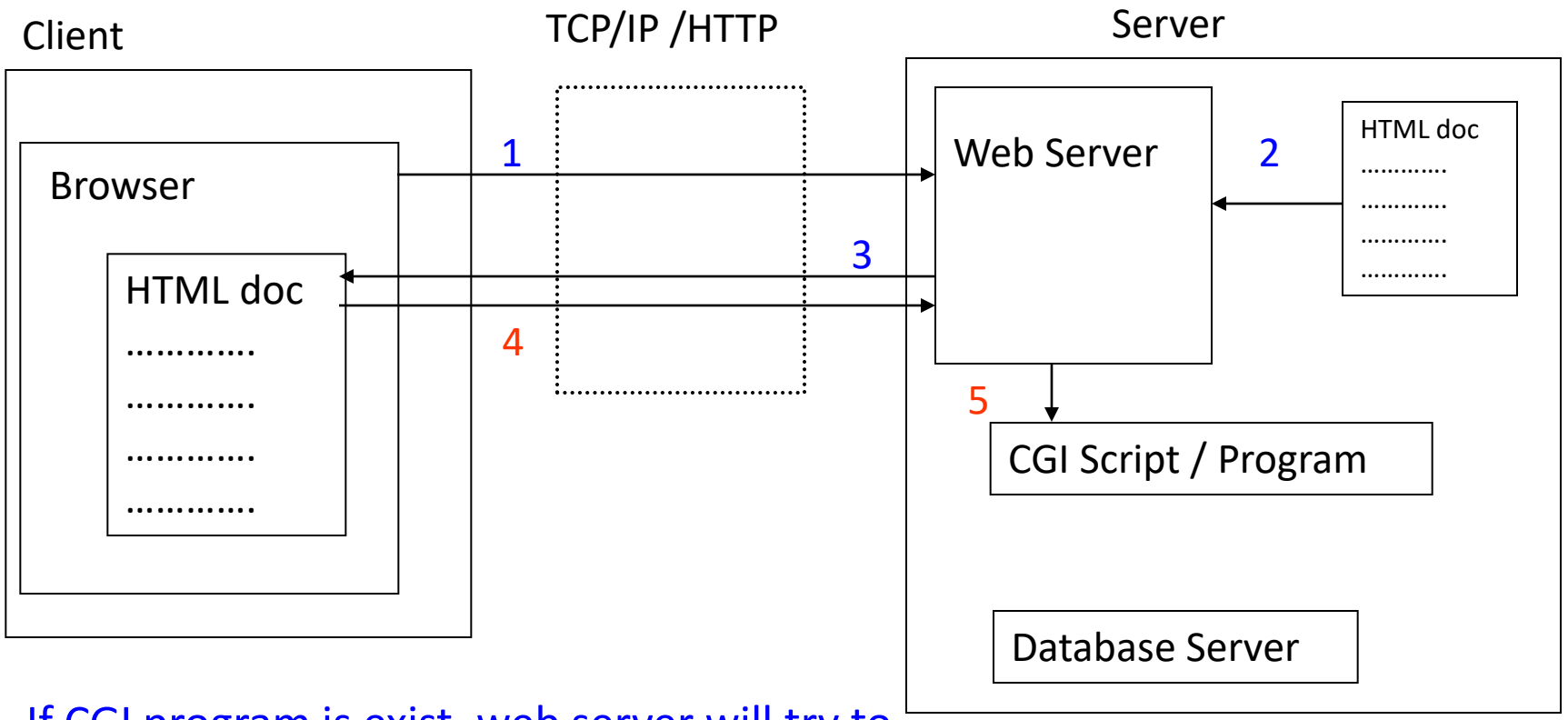
3. Web server send the HTML document to the client browser

Common Gateway Interface



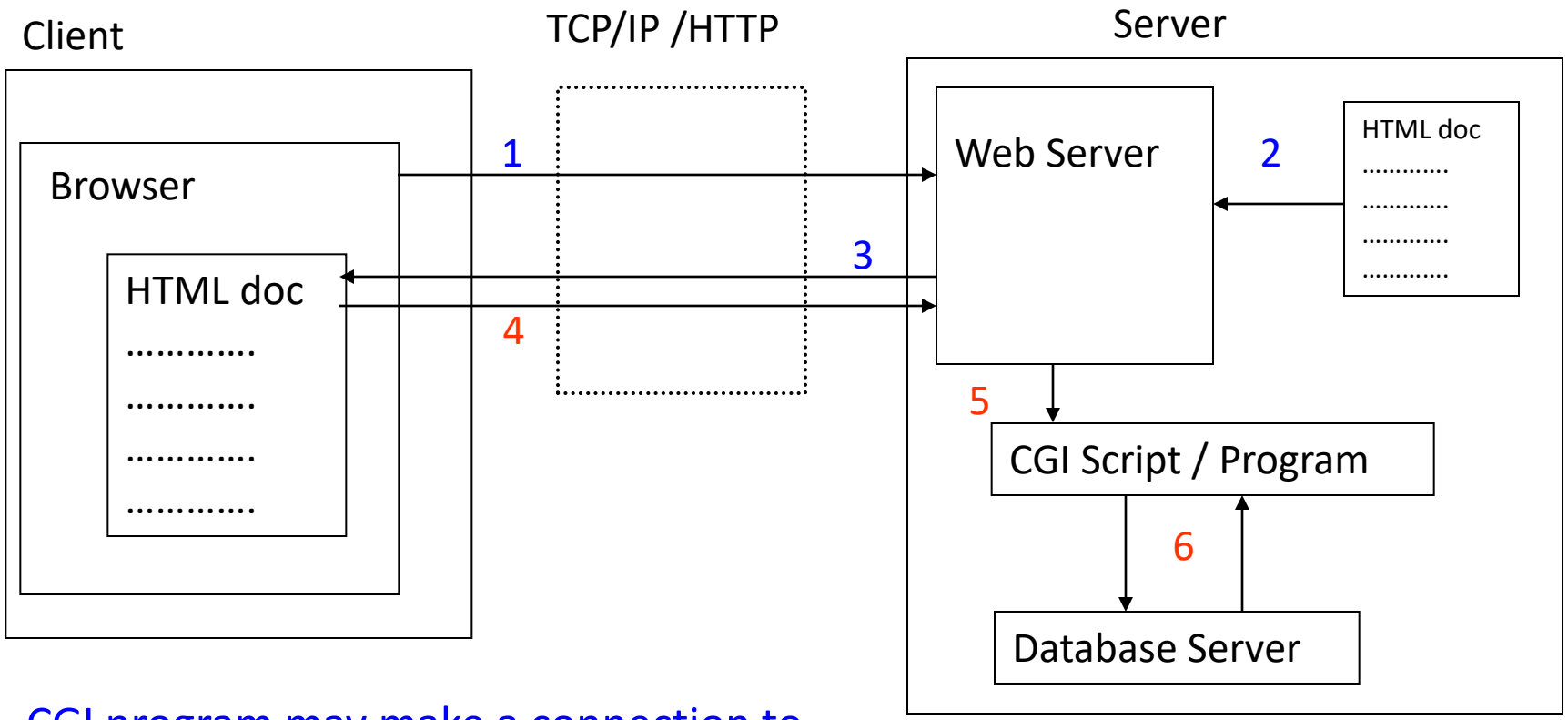
4. User may **click a submit button** in the HTML form that make a call to web server to run CGI program at server directory

Common Gateway Interface



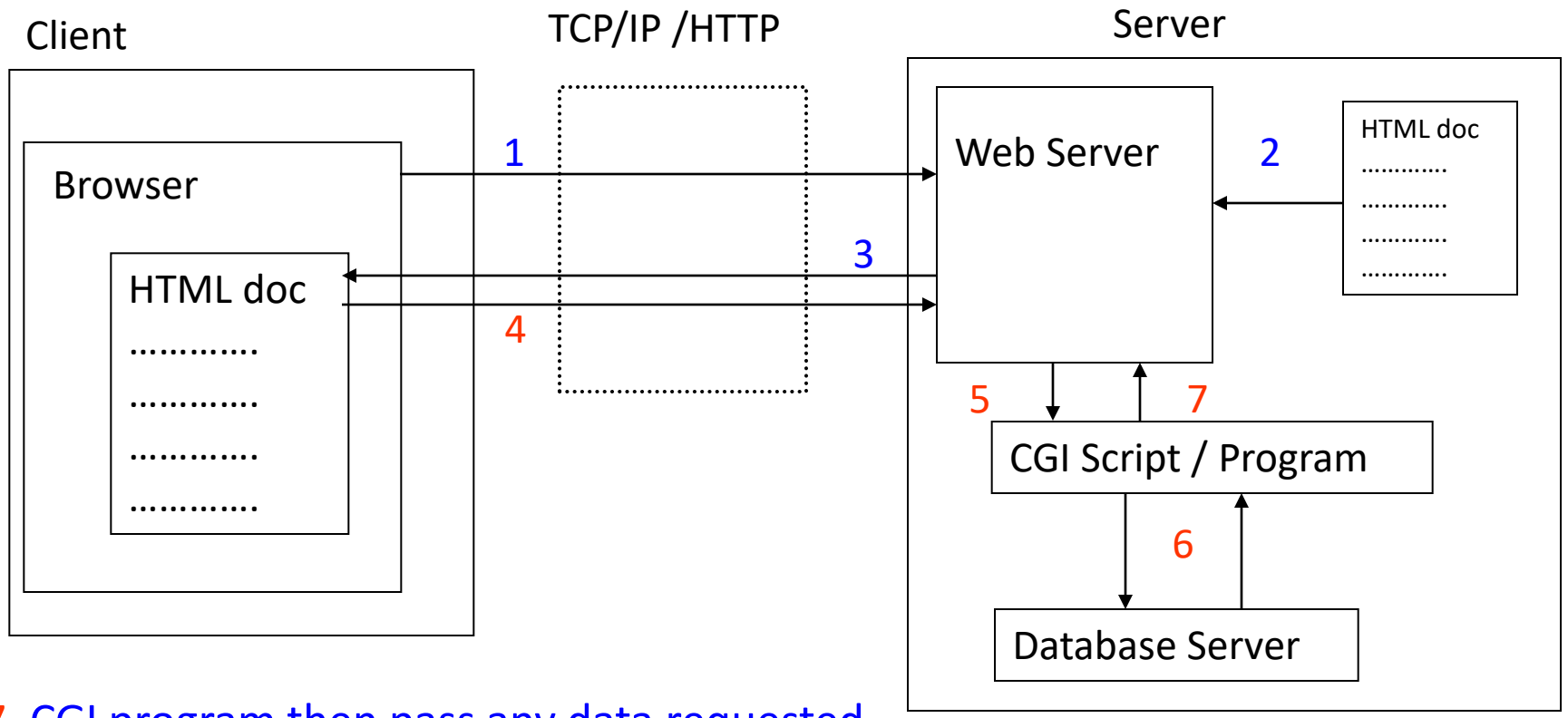
5. If CGI program is exist, web server will try to invoke the OS to run it

Common Gateway Interface



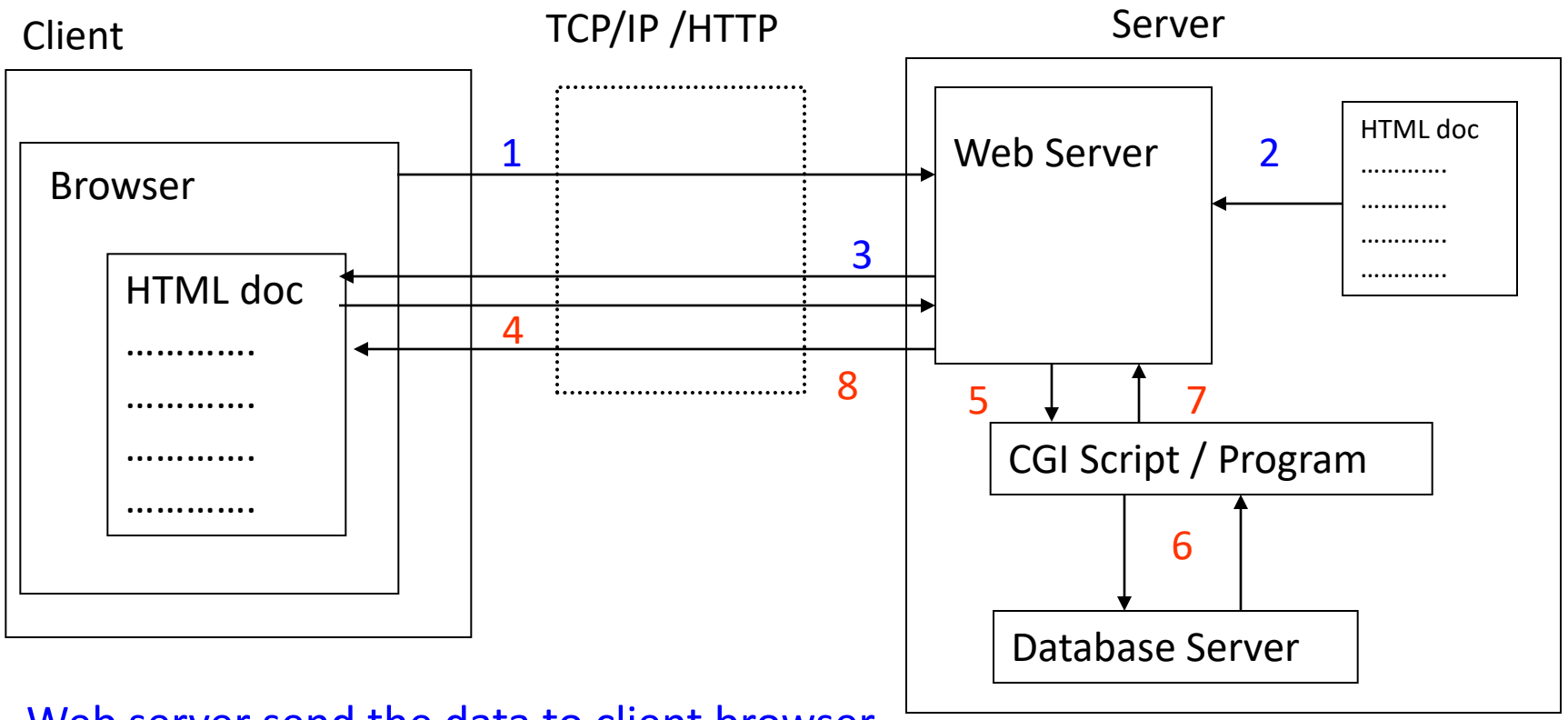
6. CGI program may make a connection to database application to retrieve data

Common Gateway Interface



7. CGI program then pass any data requested by user to the Web server

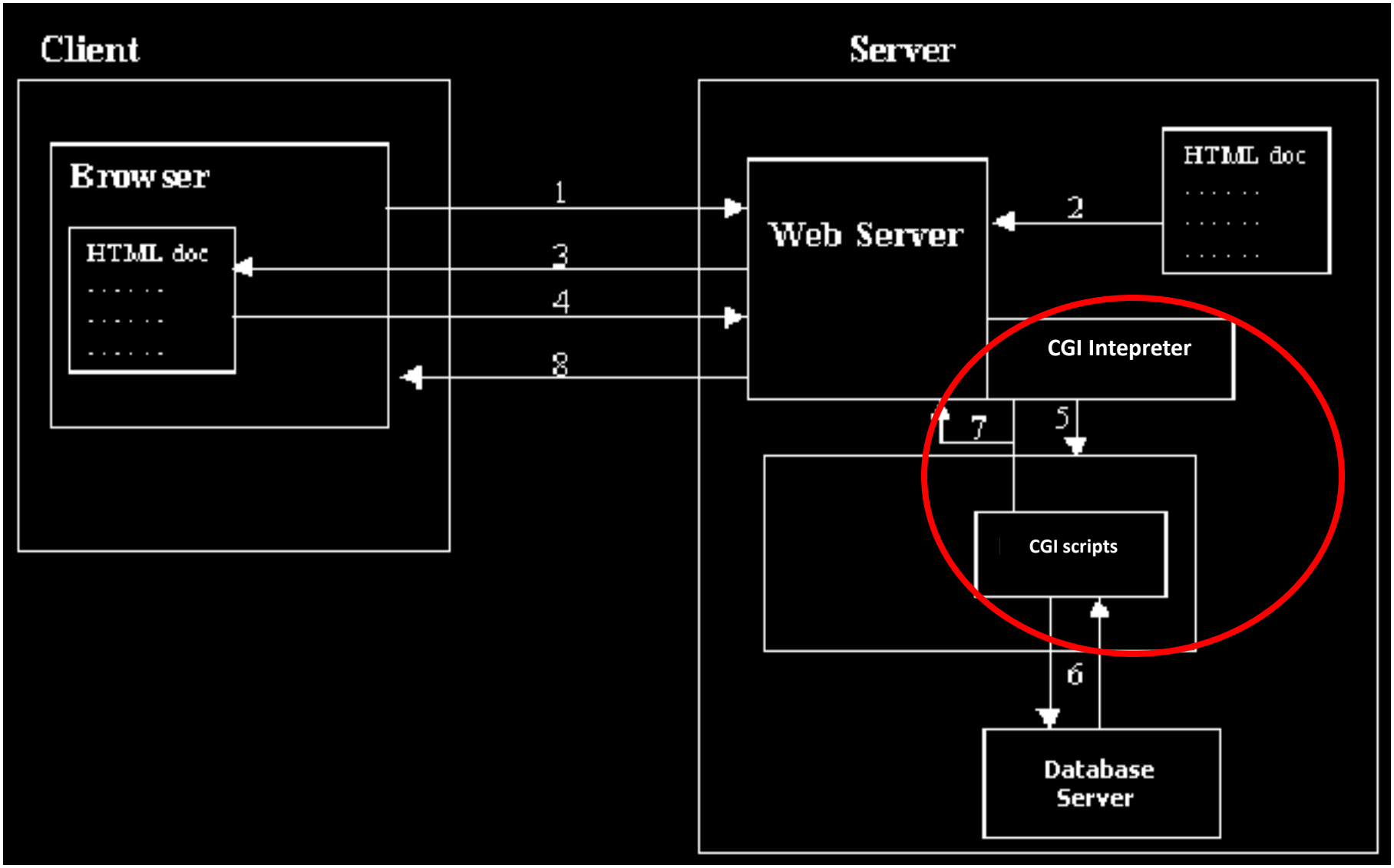
Common Gateway Interface



8. Web server send the data to client browser

Common Gateway Interface (cont.)

How the Web works for dynamic contents



CGI Programming

- CGI programs can be written using any programming language
- The languages can be categorized into two types: Server-Pages and Conventional CGI
- **Server-pages** languages:
  - The CGI codes are **embedded into HTML**
  - *eg. PHP, JSP and ASP*
- Conventional CGI:
  - using **common programming languages**
  - *eg. C, Pascal, Perl, Python, etc.*

CGI Programming (cont.)

- HTML and CGI

  Accessing cgi program through a form

```
<form method="POST" name="form1" action="login.php">
        ……..
                ……..
</form>
```

Accessing cgi program through an URL

eg. ***http://comp.fsksm.utm.my/view.php? id=1235&order=ASC***

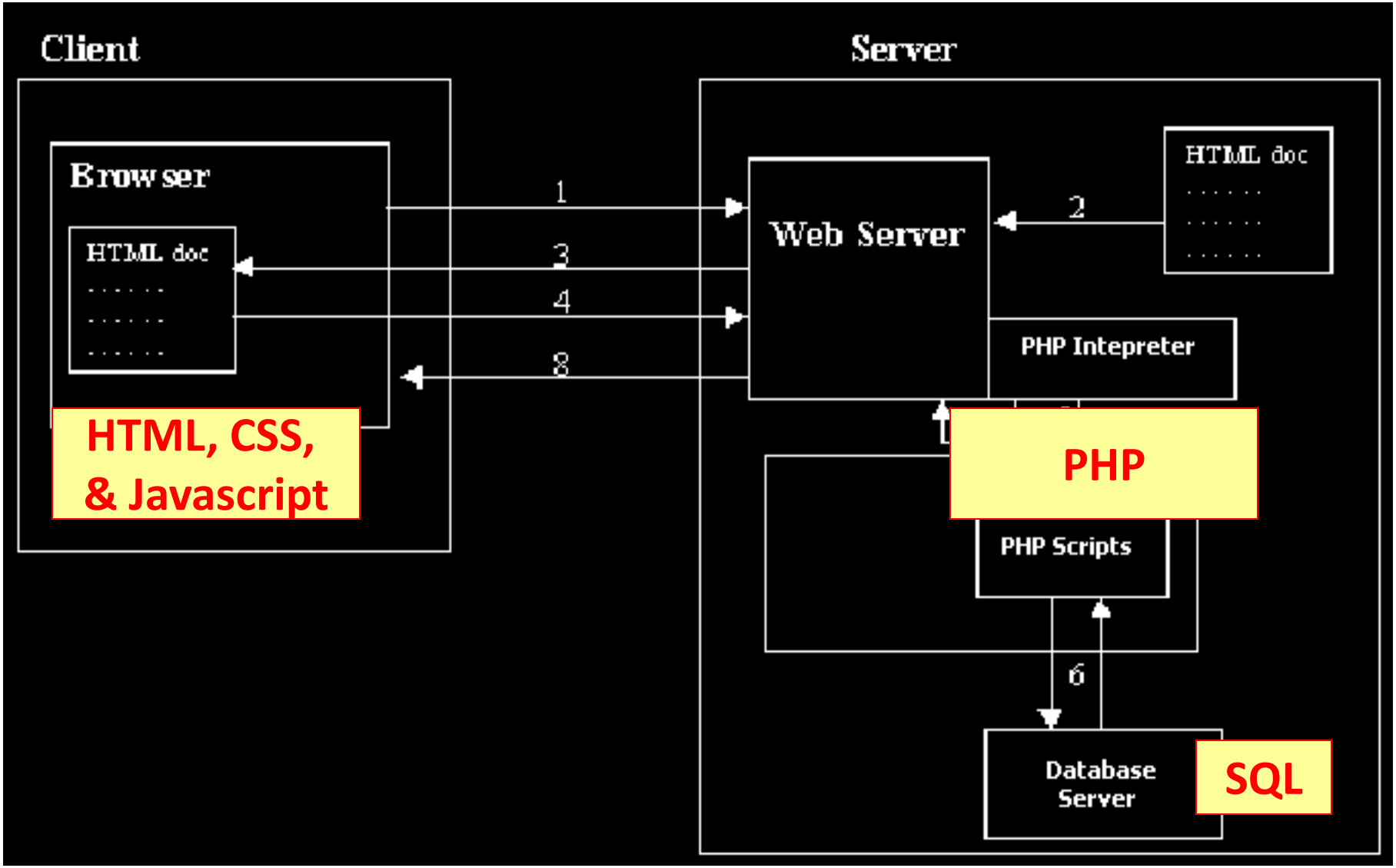
HTML vs. Javascript vs. CGI

- HTML
  - is the main language for delivering document over the web.
  - is the only language that web browsers can understand.

- Javascript
  - Documents that are created by HTML are static.
  - HTML does not support logical features.
  - Javascript is used to make the document dynamic.
  - Eg. before we can perform any delete operation, the browser should ask the user first for confirmation. This cannot be done by the HTML. Javascript can.

- CGI
  - Javascript is executed at client computers – ie. in web browsers. It is called client-side scripting language.
  - Databases or other resources are located at the server computer. We need another programs to access them – ie. CGI programs.
  - CGI languages such as PHP, JSP, Perl, etc are called server-side scripting languages, because they are executed at the server.

Our objectives …



Clients vs Servers

Clients (Browser)

- Internet Explorer
- Firefox
- Mozilla
- Netscape
- Opera
- Amaya
- AOL
- MSN

Servers

- Apache
- IIS (Microsoft)
- Netscape
- zeus
- AOLserver
- AV
- JavaWebServer
- Oracle

Scripts: Server-Side VS Client-Side

- Server-side
  - the first type possible on the Web
  - **action occurs at the server**

- Client-side
  - generally easier to implement
  - may be prepared and implemented offline
  - **action occurs on the client side (browser)**

Client-Side Scripting

- Client side scripts are **embedded inside HTML document**. They are interpreted by browser.
- When Web browser encounters a script, it calls a scripting interpreter, which parses and deciphers the scripting code.
- **Provide response** to questions and queries without interventions from the server
  - Validate user data
  - Calculate expressions
  - Link to other applications

Client-Side Scripting

Client side advantages

- Faster response time
- Better animation
- Simpler server programs

Client side disadvantages

- Longer load time
- Browser compatibility
- Complexity in web page design

Client-Side Scripting: JavaScript

- A scripting language for Web pages, developed by Netscape in 1995
- JavaScript code is embedded directly in HTML (interpreted by browser)
- good for adding dynamic features to Web page, controlling forms and GUI

Advantage

- Easy to learn and use
- Wide browser support
- Protection of local resources

Disadvantage

- Browser compatibility issues
- Not object oriented
- Unable to gain access to local resources

Client-Side Scripting: Java Applet

- Can be server-side or client-side
- Can define small, special-purpose programs in Java called applets
- Provides full expressive power of Java (but more overhead)
- Applets are included in Web pages using special HTML tags
- Interpreted by the Java Virtual Machine embedded in the browser
- Good for more complex tasks or data heavy tasks, such as graphics

Client-Side Scripting: Java Applet

Advantage

- High functionality
- Object oriented and full graphics functionality
- Protection of local resources
- Wide Browser support
- With Java2, be able to gain access to local resources with signed applets

Disadvantage

- JVM(Java Virtual Machine) compatibility issues
- Difficulty to install and configure for local access
- Loading time and performance may be poor for large application.

Server-Side Scripting

Advantages

- Allows creation of dynamic web pages
- Modifies HTML code on the server before sent to client
- Uses databases such as Access, MySQL and Oracle
- Responds to user input

Disadvantages

- More complicated then HTML (with debugging)
- Slower to load on the server
- Harder to learn
- Web server must be enabled

Examples of Server-Side Scripts

- CGI (Common Gateway Interface)
  - A standard for interfacing external applications with information servers, such as HTTP or Web servers
  - CGI program is any program designed to accept and return data that conforms to the CGI specification
  - CGI program can be written in any language that allows it to be executed on the system, such as: C/C++, Fortran, PERL, TCL, Any Unix shell, Visual Basic, AppleScript etc.
- PHP (*H*ypertext *P*reprocessor)
  - Widely-used Open Source general-purpose scripting language that is especially suited for web development and can be embedded into HTML.

Examples of Server-Side Scripts

- Active Server Pages (Microsoft)
  - ASP is a Microsoft Technology and run on IIS (Internet Information Server) & PWS (Personal Web Server)

- Java Server Pages (Sun)
  - Sun's solution for developing dynamic web sites
  - JSP enable the developers to directly insert java code into jsp file, this makes the development process very simple and its maintenance also becomes very easy

Summary

- We will learn in this class how to:
  - present our content for web viewing in its most simple way through (HTML code)
  - systematically managing the formatting and arrangement of content in HTML through CSS
  - managing event, interaction and manipulating web content (html doc) through Javascript
  - management of user and application data through server-pages scripting (server code encoded in html doc)

YouTube Video

- http://www.youtube.com/watch?v=gY5hCB8f7wk&feature=related
- http://www.youtube.com/watch?v=jJoodliuy5I&feature=fvwrel